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MIT SCHOOL OF COMPUTING

Class:
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DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

BHUNAGA

A Multilingual Retrieval-Augmented Advisory System for Indian Farmers

Manish Yadav, Aishwarya Deshmukh, Harshal Agre, Jayesh Kapade

Faculty Guide:
Prof. Minakshi Sonwane

Farmer Says:
- "I don't know today's mandi price"
- "Pests are ruining my crop"
- "I can't understand English apps"
- "Only the pesticide shop advises me"

Farmer Thinks:
- Wish everything was in my language
- Right advice would save me money
- Good to know weather in advance
- I need information I can trust

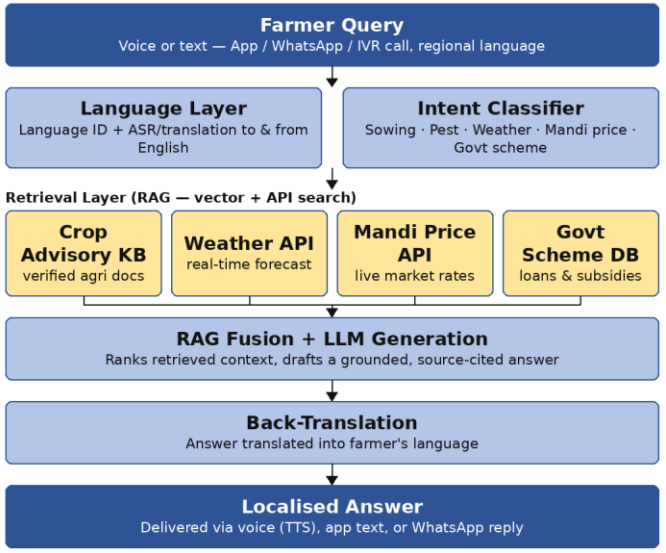
Farmer

What Farmer Does:
- Asks the local pesticide retailer
- Waits for TV/radio weather news
- Travels to the mandi to check rates
- Copies what other farmers do

What Farmer Feels:
- Anxious about losing the crop
- Frustrated by the language barrier
- Unsure about generic chatbot answers
- Hopeful for a simple own-language tool

Sr. No.	Requirement	Proposed Solution
1.	**Crop & Pest Advisory**	
	Available Solutions- - Local pesticide retailers - Generic AI chatbots	**Proposed Solution-** - RAG pipeline over verified agri-advisory sources, in the farmer's own language
2.	**Weather Updates**	
	Available Solutions- - TV / radio forecasts - English-only weather apps	**Proposed Solution-** - Real-time weather feed delivered in local language
3.	**Mandi (Market) Prices**	
	Available Solutions- - Manual mandi visits - Scattered govt. portals	**Proposed Solution-** - Live mandi price retrieval via voice or text query
4.	**Language Access**	
	Available Solutions- - English-only agri apps - Unreliable general chatbots	**Proposed Solution-** - Multilingual voice & text NLP interface for Indian regional languages

Block Diagram



Problem statement

Farmers in India plan crops in **their own language,** but advice on sowing, pests, weather and mandi prices is mostly **in English** and scattered online, so many rely on local pesticide sellers **who** push unneeded chemicals. What's missing is a trusted system giving accurate, localised advice in the farmer's own language.

Proposed Solution

Bhunaga is a multilingual, retrieval-augmented advisory system where farmers ask sowing, pest-control, weather and mandi-price questions in their own language and get accurate, localised answers grounded in trusted agricultural sources.

Scope and Feasibility

4 Modules:

1. Multilingual voice & text interface
2. Retrieval-augmented advisory engine
3. Real-time weather & mandi price feeds
4. Regional language crop & pest advisory